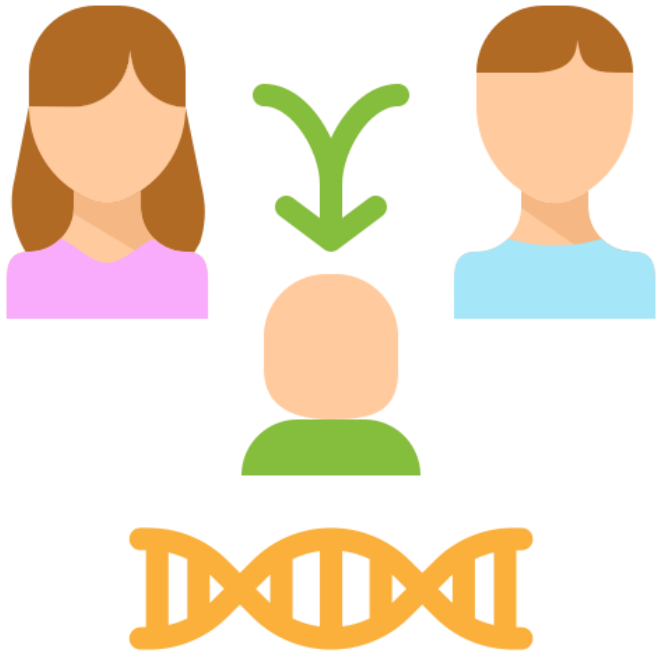
Genetics and Heredity

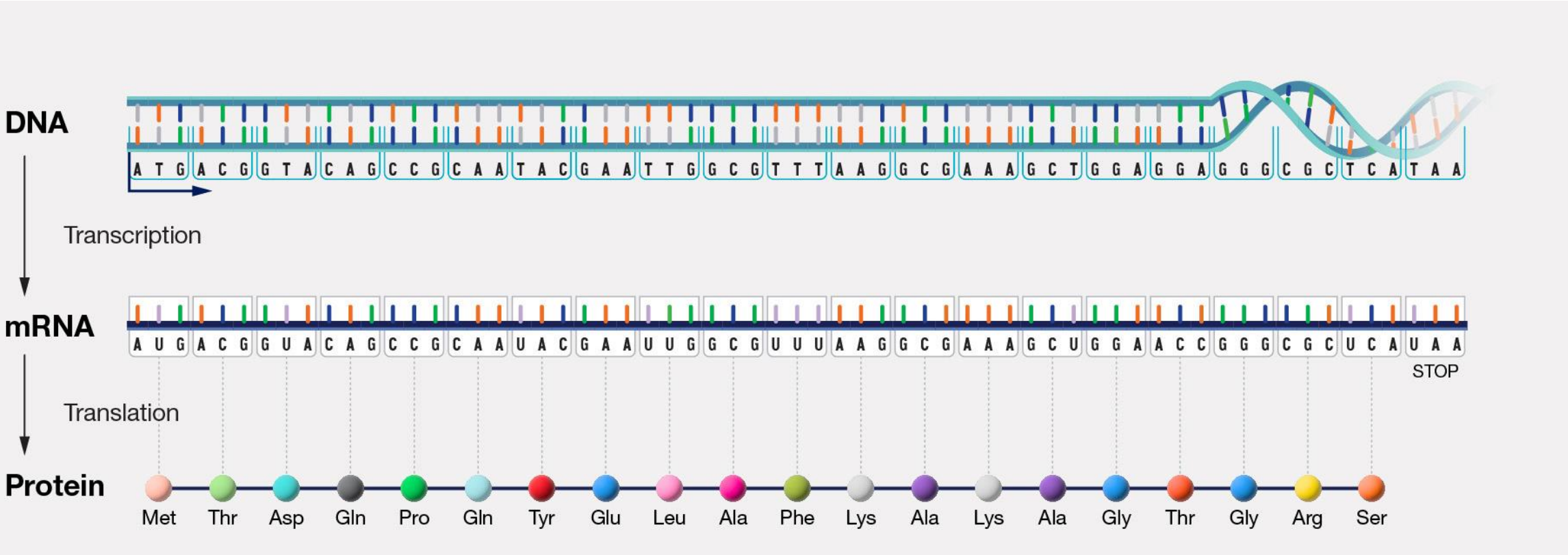
Lecture 06



Heredity and traits

- We are who we are and (although debatable) we do what we do is because of this 1.8m of DNA in each of our cells
    - Our DNA
    - Our genome
- Human genome consists of approximately 20,000 genes
- Each of these genes is the instructions for building a single protein
    - And these proteins, when built, carry out ALL of the cellular processes that keeps us alive
- In this way, DNA is quite literally the "instruction manual of life"
    - But it all comes down to proteins, protein are the machines
- So a single mutation in a single gene only affects a single protein

Central Dogma of Molecular Biology



https://www.genome.gov/genetics-glossary/Central-Dogma

Deciphering the Genetic Code

- 1865 – Mendel defined the basic unit of inheritance as the gene
- 1900 – Mendel's forgotten work resurfaces; nature of gene is still unknown
- 1902 – Boveri and Sutton comes up with chromosome theory
- 1944 – it is established that a gene is made of DNA
- 1952-53 – Watson-Crick's double helix structure for DNA

DNA: = {A, C, G, T}

RNA: = {A, C, G, U}

Double Stranded DNA

5' A T T G C C C A T 3'

| | | | | | | | | |

3` T A A C G G G T A 5'

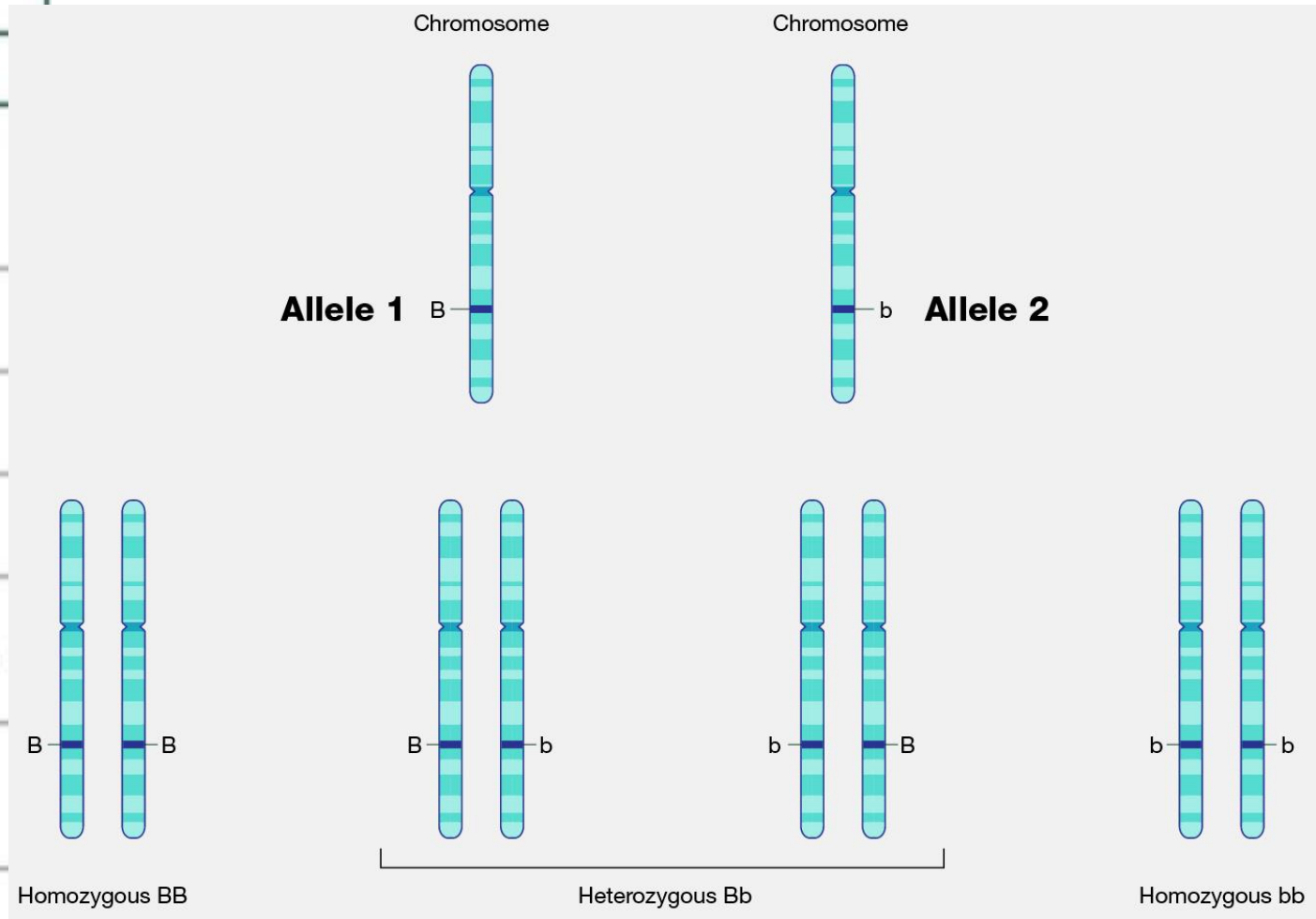


https://www.youtube.com/shorts/JpJEEGuc0zI

Some terminology

Table 3.1 Summary of important genetic terms

Term	Definition
Gene	A genetic factor (region of DNA) that helps determine a characteristic
Allele	One of two or more alternative forms of a gene
Locus	Specific place on a chromosome occupied by an allele
Genotype	Set of alleles possessed by an individual organism
Heterozygote	An individual organism possessing two different alleles at a locus
Homozygote	An individual organism possessing two of the same alleles at a locus
Phenotype or trait	The appearance or manifestation of a character
Character or characteristic	An attribute or feature

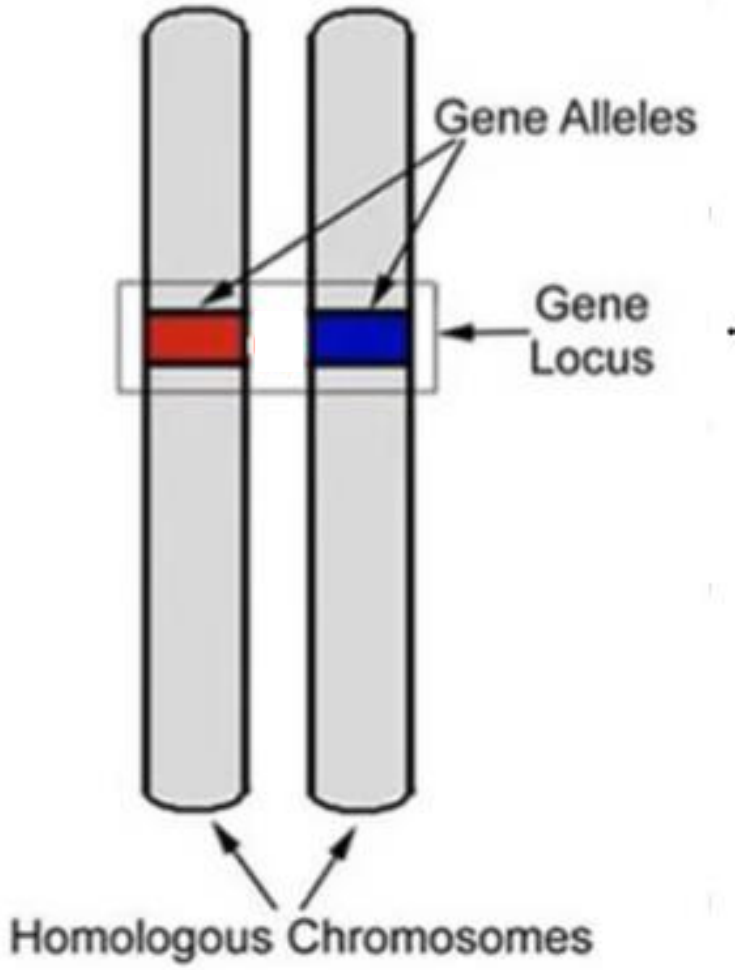


https://www.genome.gov/genetics-glossary/Allele

Some terminology

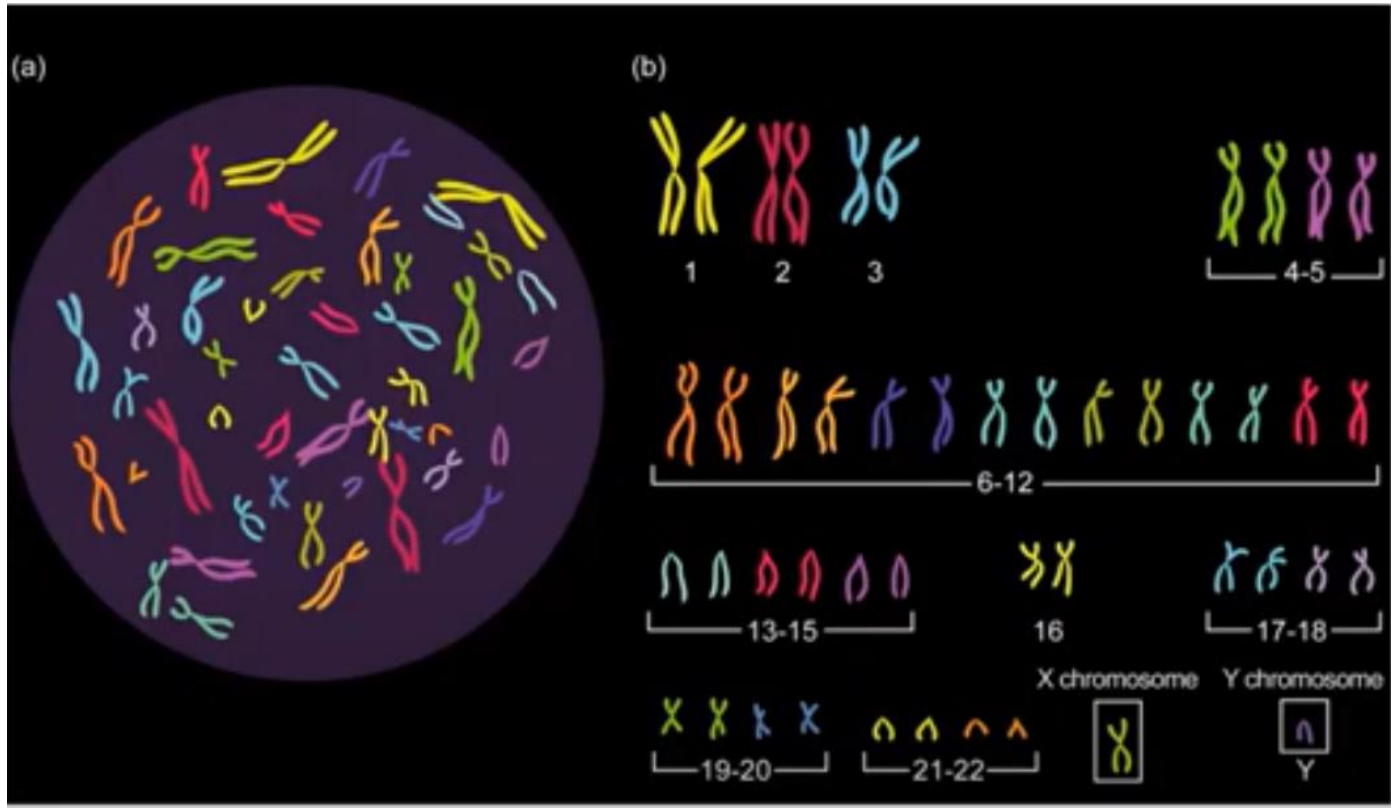
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Heredity and traits

- You get two genes for each trait – one from mother and another from father
  - If both genes are “wildtype”, you show the ‘normal’ trait
  - If one gene is “wildtype”, you *generally* show the ‘normal’ trait
  - If neither gene is ‘wildtype’, you show a different trait



Some terminology

Table 3.1 Summary of important genetic terms

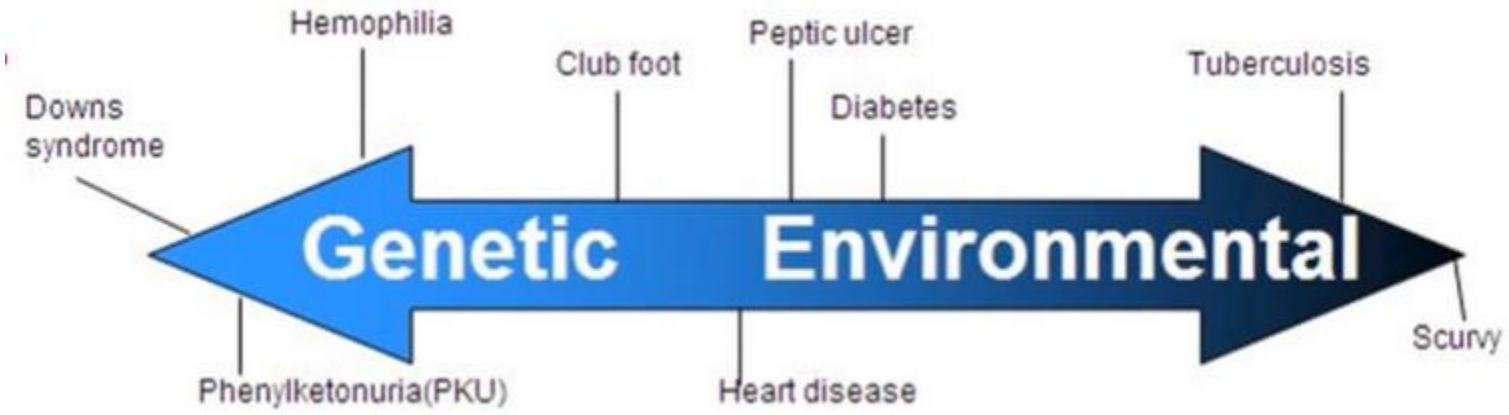
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- A genotype is an organism's entire set of alleles
  - It is sum total of organism's entire genetic information
- A phenotype is the physical expression of the organism's genotype
  - E.g. someone's mother has blue eyes and father has brown eyes.
  - The genotype is blue/brown
  - Phenotype is the actual color of the eyes
- **Dominant** and **Recessive** gene: An allele of a gene is said to be dominant when it **effectively overrules the other** (recessive) allele

Some terminology

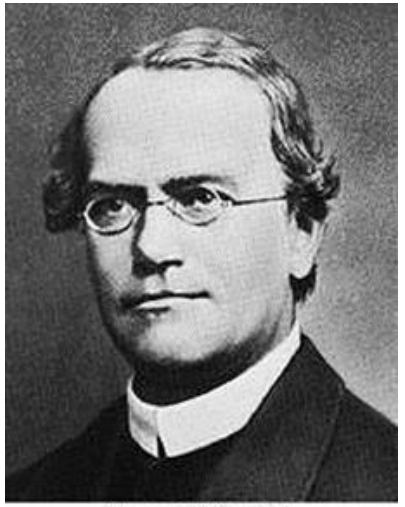
- Anything can be a phenotype
  - Hair color, skin color, eye color etc
  - The likelihood to develop breast cancer
  - A cell's ability to repair it's DNA

- Genes only get you so far
  - The environment plays a huge role in phenotype as well
  - Nature vs. nurture
  - A kid can have alleles of tall parents, but if he does not receive proper nutrition, he may not grow tall.



https://notthroughignorance.wordpress.com/tag/phenylketonuria/

Mendelian Genetics

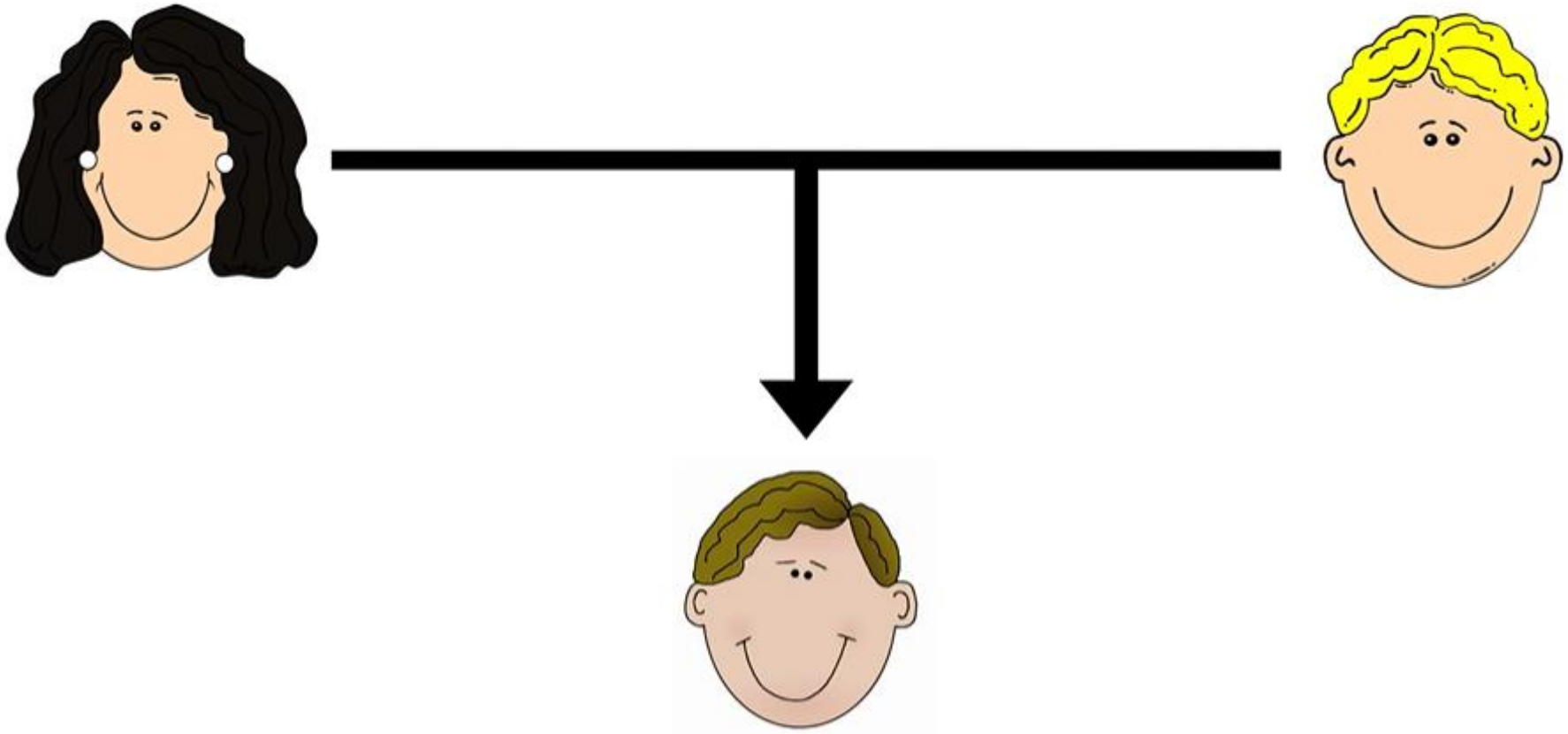

Gregor Mendel

- An Austrian monk, Gregor Mendel, was the first person to methodically study and understand how traits are passed from parents to offspring.

- He was able to understand the concept of heredity long before we knew cells or DNA existed.

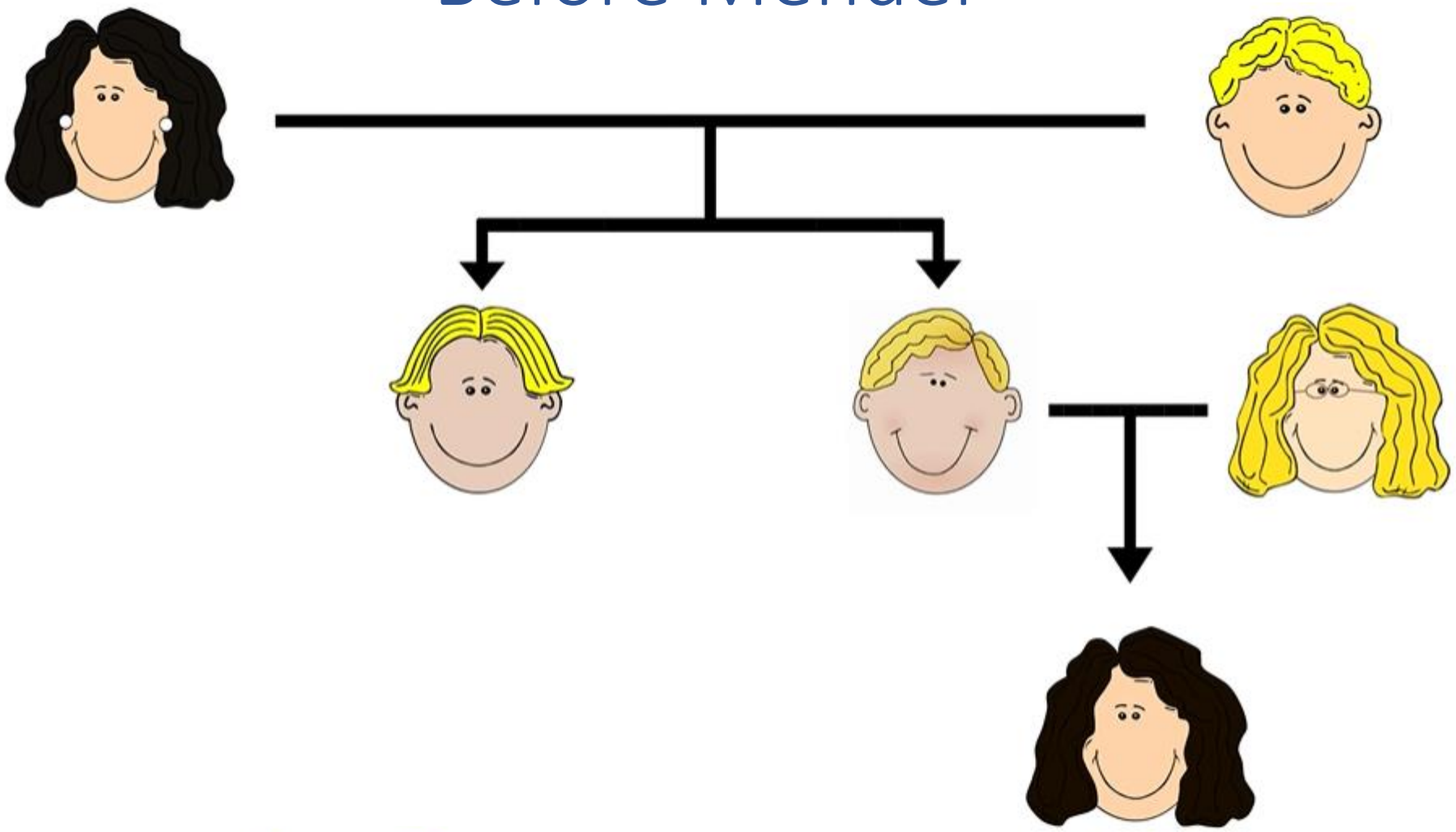
- What he did was controlled crosses
  - He took parents that looked very different, mated them and then observed the appearance of the offspring
  - He used pea plants as his model organism

Before Mendel



Heredity was believed to involve a blending of attributes – Wrong notion

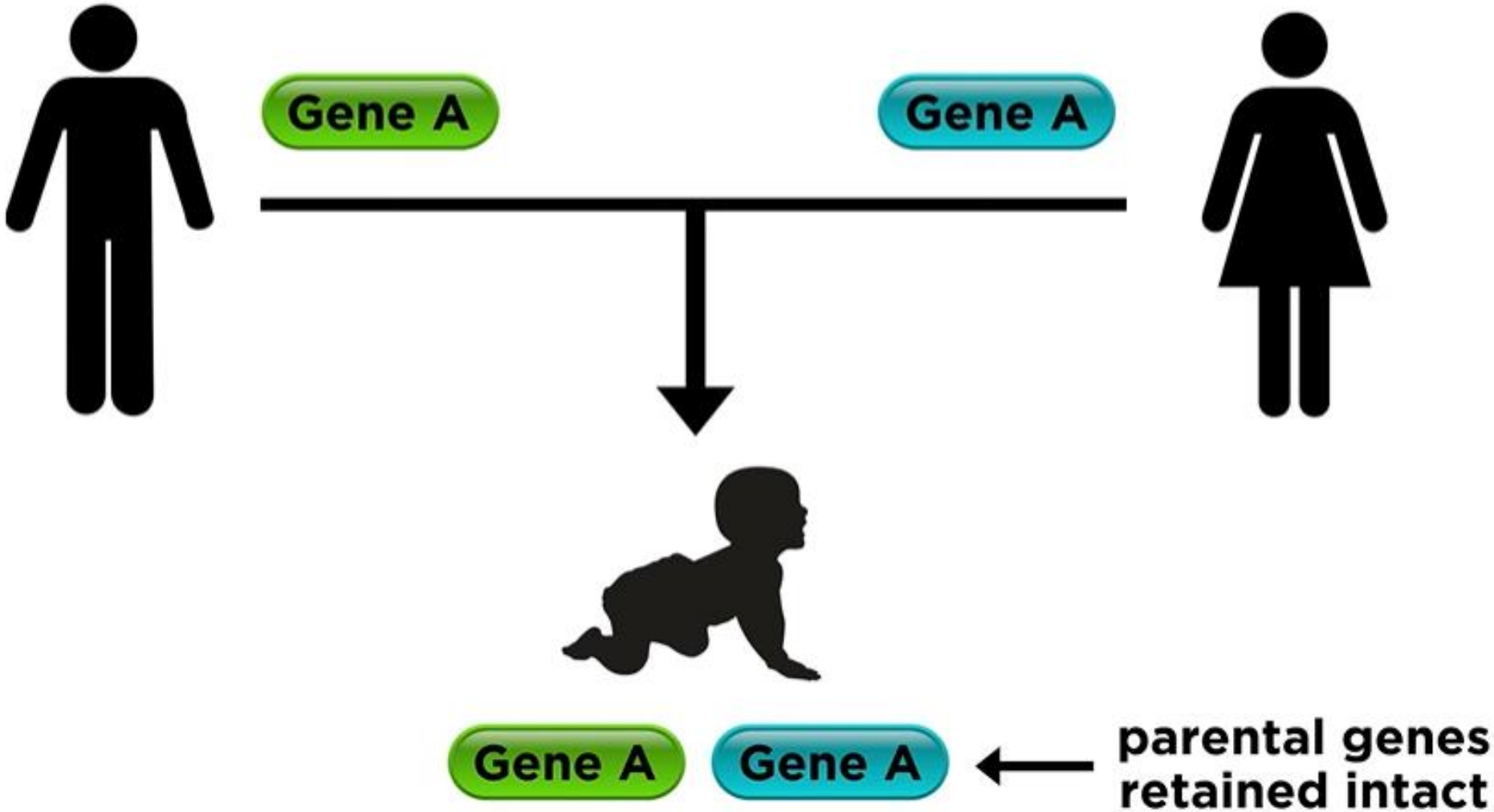
Before Mendel



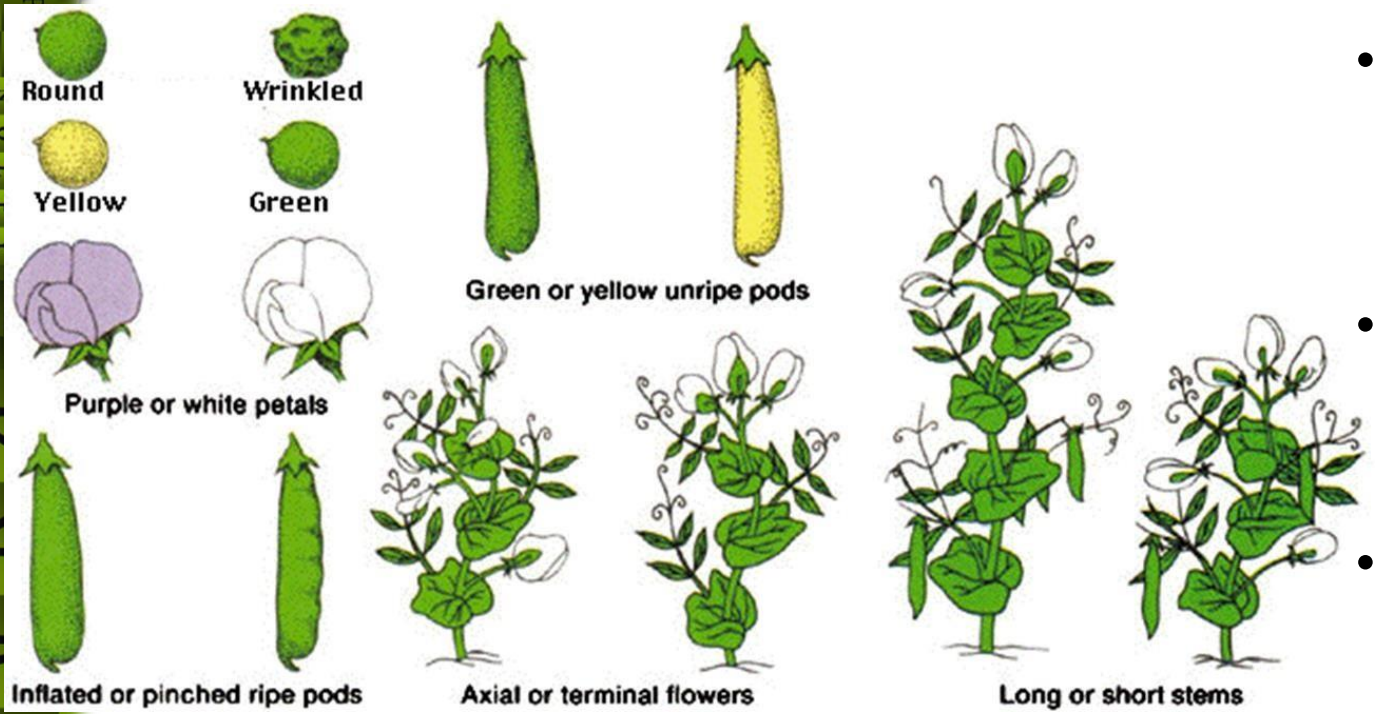
This did not account for the way traits can skip a generation!

Heredity must be based on discreet units of inheritance.

Gene theory – each set of gene remains intact in the offspring from parents



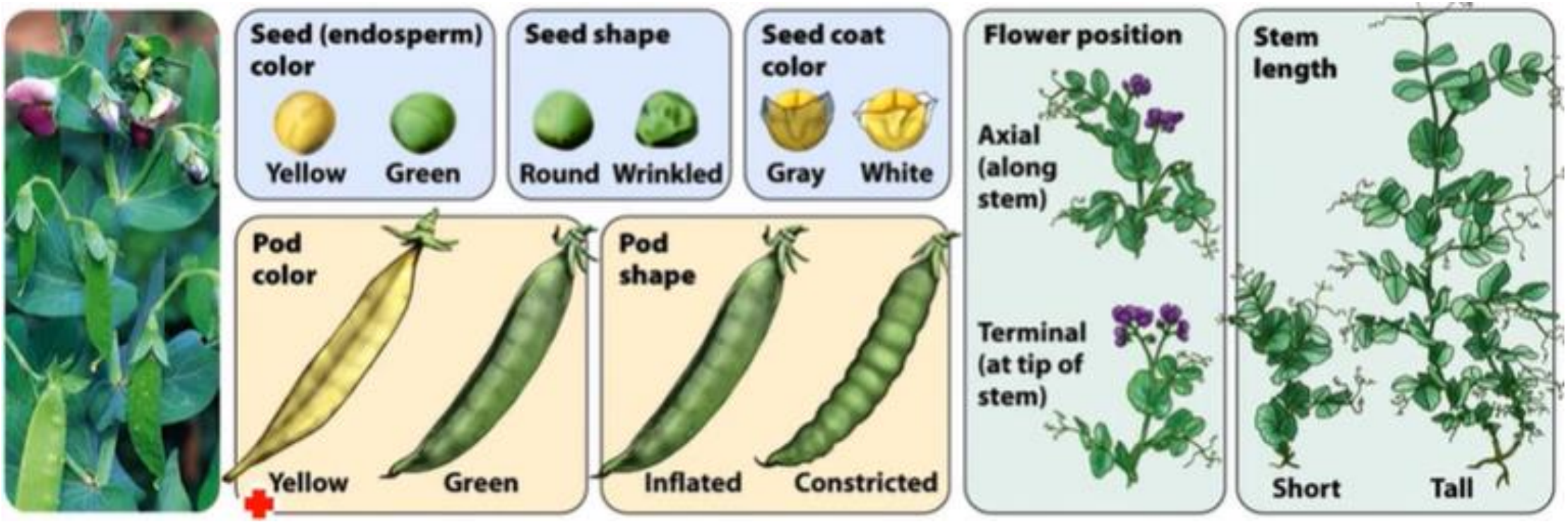
Mendelian Genetics



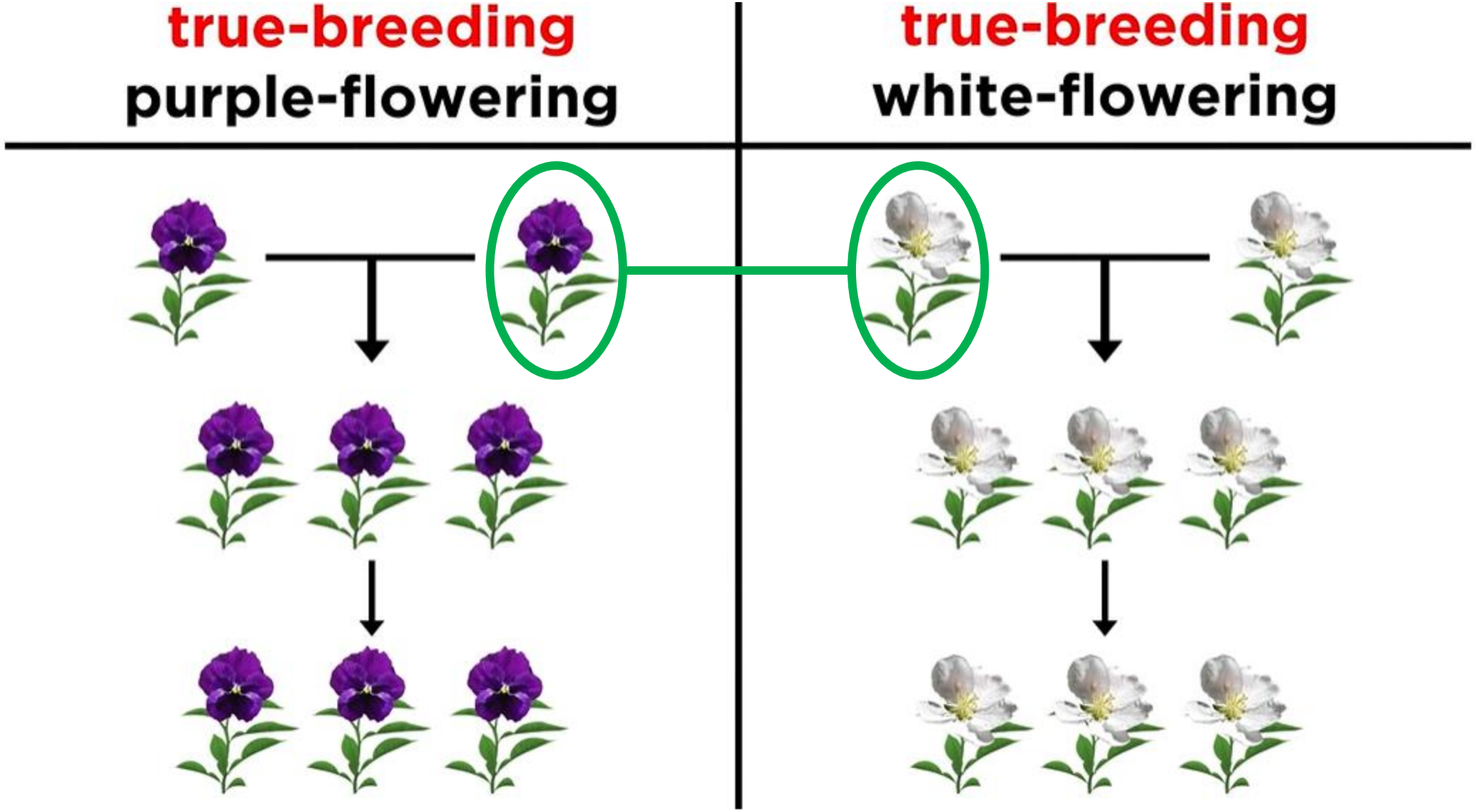
- Mendel probably chose to work with peas as they are available in many varieties.
- He used pea plants as their short generation time and the large number of offspring.
    - Excellent for statistical analysis
- Mendel chose to track only those characters that occurred in two distinct, alternative forms.
- A heritable feature that varies among individuals, such as flower colour, is called a character.
- Each variant for a character, such as purple or white colour for flowers, is called a trait.

Mendelian Genetics

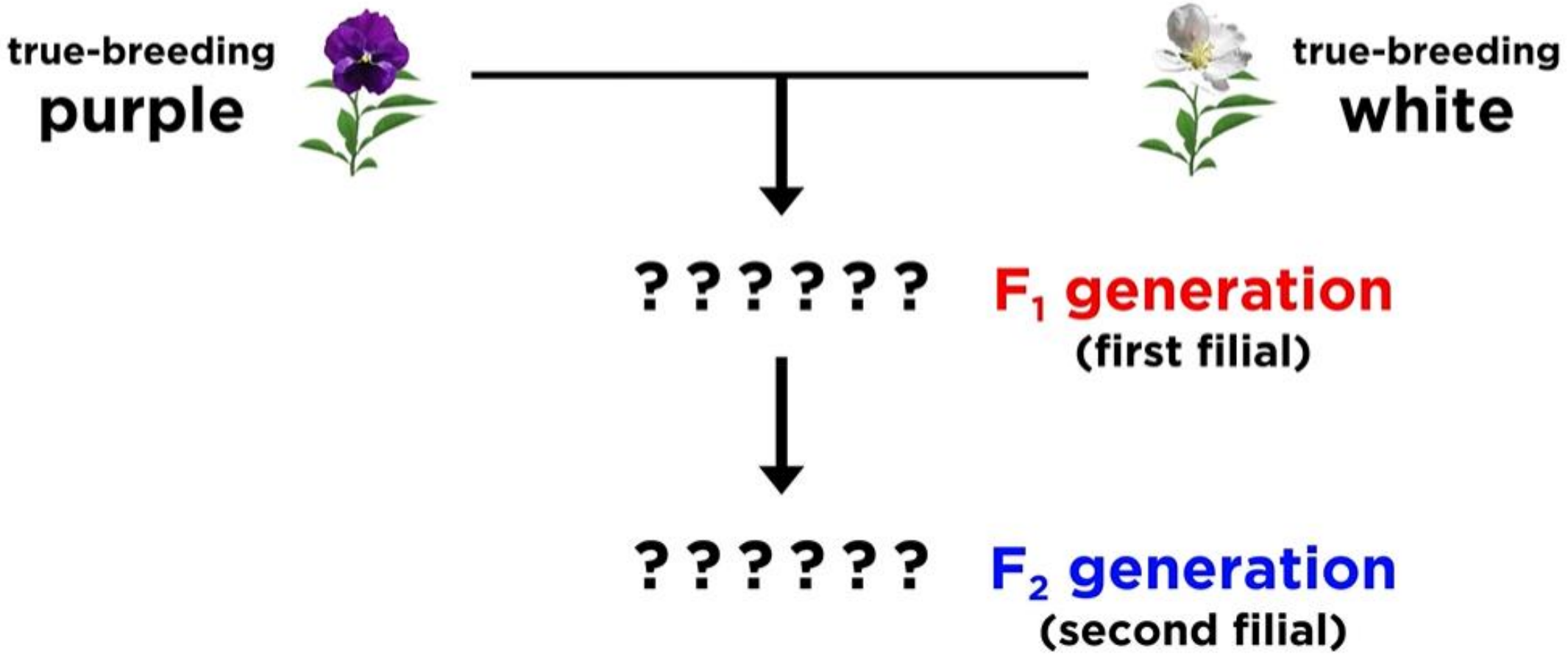
Mendel chose to study seven different characteristics of pea plants



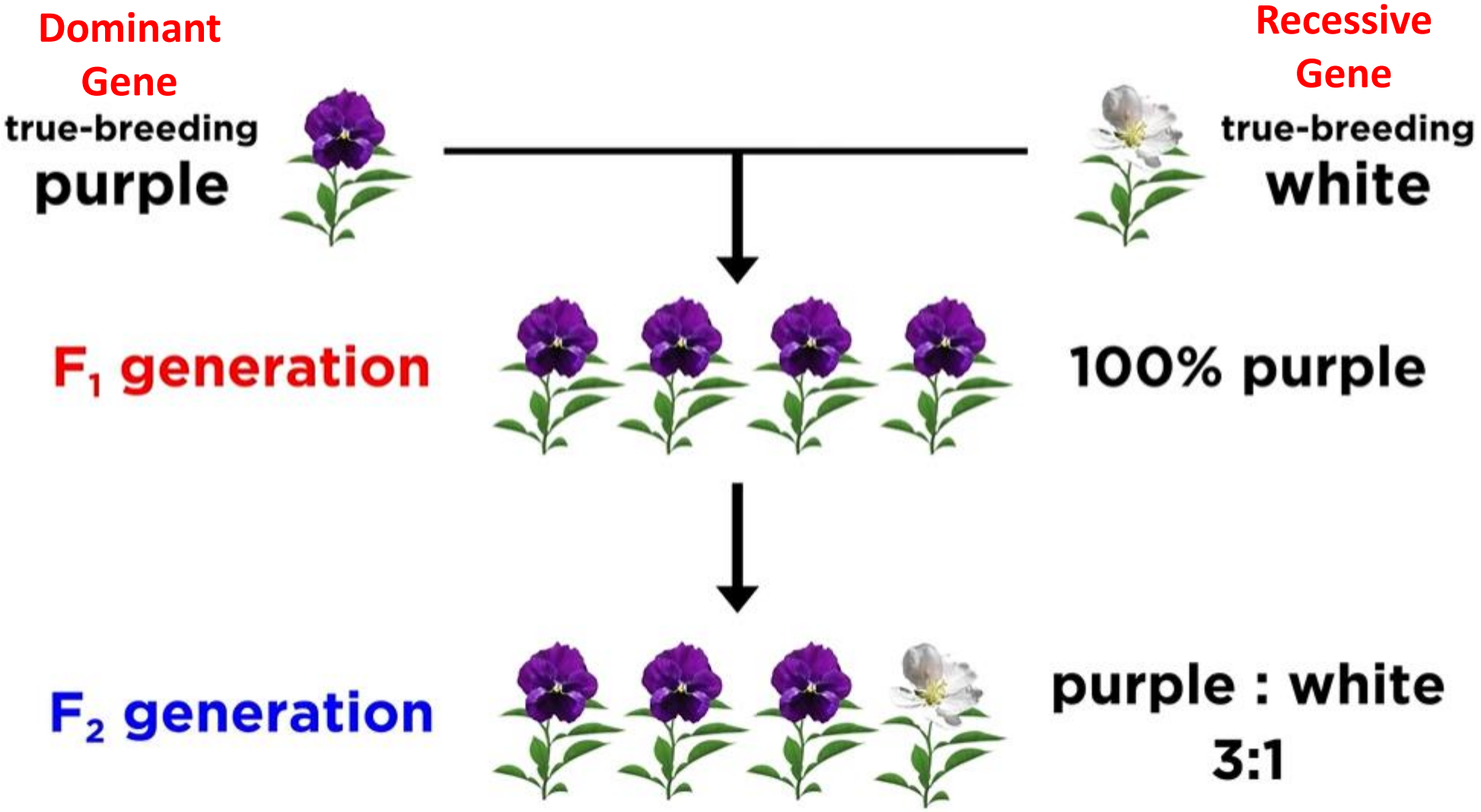
Mendelian Genetics



Mendelian Genetics



Mendelian Genetics



Information for white-flowering plant was not lost in the F1 generation